



ZHANG JIANNAN

Flight Control Engineer & PhD Candidate

@ jiannan.zhang@tum.de

+49-176-630-81506

Munich, Germany

jiannan-zhang

STRENGTHS

Flight Control System

Flight Physics Modelling

MBSE

Certification

Flight Testing

SOFTWARE SKILLS

MATLAB/Simulink

Control Design Toolbox

Math Symbolic Toolbox

Git

C/C++

LANGUAGES

English: Professional

Chinese: Native

German: Goethe B1

SUMMARY

Motivation: Productize eVTOLs.

Industry: 10+ years on eVTOL FCS, 1 TC, 1000+h flight testing.

Academy: 10+ publications, 300+ citations.

PROFESSIONAL EXPERIENCE

Research Associate | Institute of Flight System Dynamics, TUM

06.2016 – Now

Munich, Germany

- Lead of control allocation algorithm.
- Lead of control configuration optimization.
- Flight control system development.
- Flight dynamics modelling toolchain development.

Lead of Flight Control System | AutoFlight

08.2023 – 07.2024

Shanghai, China

- Lead of GNC function development.
- Lead of system architecturing.
- Lead of FCS TC applications.

Senior Flight Control Engineer | AutoFlight

07.2021 – 08.2023

Shanghai, China

- Lead of guidance and control law development.
- Lead of flight dynamics modelling development.
- System architecturing.
- Certification tasks (requirement capture and validation).
- Flight testing.

Flight Control Engineer | Autel Robotics Europe GmbH

06.2016 – 01.2020

Ismaning, Germany

- Flight control algorithm for tilt-rotor UAV (Dragonfish).
- Attitude loop design and control allocation.
- Flight testing.

Software Developer | Phoenix-Wings GmbH

06.2020 – 07.2021

Ismaning, Germany

- Lead of trim and linearization tool.
- Simulation toolchains development.
- Flight testing.
- Project management.

EDUCATION

Doctoral Candidate [Technical University of Munich \(TUM\)](#)

📅 11.2017 – Now (Expected Grad.: 2026) 📍 Munich, Germany

- **Research Direction:** eVTOL Control Authority and Allocation Optimization

M.Sc. Aerospace Engineering [Nanyang Technological University \(NTU\)](#)

📅 08.2013 – 02.2016 📍 Singapore

- **CGPA:** 4.92 (CAP: 1.06) **Major Subjects:** Flight Control, Light Weight Structures

B.Eng. Aircraft Design Engineering [Beijing Institute of Technology \(BIT\)](#)

📅 08.2008 – 07.2012 📍 Beijing, China

- **GPA:** 85/100 **Major Subjects:** Flight Mechanics, Aerodynamics

INDUSTRY PROJECTS

CarryAll - 2000kg Lift-cruise eVTOL | [Autoflight](#)

📅 07.2021 – 08.2024 DEV Industry Project

- **Role:** Lead of modelling, guidance and control law
- **Contribution:** Control law design, SW implementation, system architecturing and certification tasks (FHA, PSSA, etc.).
- **Highlight:** First 2000kg eVTOL TC in the world.

V280 - 280kg Lift-cruise eVTOL | [TUM-FSD](#)

📅 02.2020 – 06.2021 DEV Industry Project

- **Role:** Lead of control allocation development
- **Contribution:** Control allocation algorithms for hover, transition and cruise flight, fail-safe proven for OEI.
- **Highlight:** Developed under code-compliant guidelines specified by DO-331 (standard libraries, Model Advisor, Test Harnesses, etc.).

Manta Ray - 30kg Lift-cruise eVTOL | [Phoenix-Wings GmbH](#)

📅 06.2020 – 07.2021 DEV Industry Project

- **Role:** Lead of Trim and Linearization Tool in C++
- **Contribution:** Implemented the trim and linearization tool in C++, and the MATLAB/Simulink interface for model verification and further design tasks.

Dragonfish - 7kg Tilt-rotor UAV | [Autel Robotics Europe GmbH](#)

📅 06.2016 – 01.2020 DEV Industry Project

- **Role:** Lead of control allocation development
- **Contribution:** Developed nonlinear control allocation for the entire flight envelope. Support INDI attitude and velocity loop design. Two first-author papers published.
- **Highlight:** Dragonfish as a product has already generated billions of sales revenues.

RESEARCH PROJECTS

Generalized Control Allocation Protection | [TUM-FSD](#)

📅 01.2025 – 12.2025 DEV Research Project

- **Role:** Lead of algorithm development
- **Contribution:** An adaptive method which generalize p-norm optimization targets. Two command protection methods to improve pseudoinverse based methods. One first-author and two second-author journals published.

eVTOL Control Configuration Optimization | TUM-FSD

📅 05.2020 – 12.2024

🔬 Research Project

- **Role:** Lead of algorithm design and SW implementation
- **Contribution:** Developed a method to optimize eVTOL powered-lift rotor topology. The optimal configuration features sublime robustness against OEI scenario. Two first-author journals published.

F-16 Modelling Extension and Control | TUM-FSD

📅 05.2016 – 01.2020

🔬 Research Project

- **Role:** Lead of modelling and INDI baseline controller
- **Contribution:** Developed the INDI closed-loop simulation framework and the RSPI nonlinear optimal control allocation method. Two first-author papers published.

PUBLICATIONS

1. *Attainable Moment Set Optimization to Support Configuration Design: A Required Moment Set Based Approach*
2. *Saturation Protection with Pseudo Control Hedging: A Control Allocation Perspective*
3. *Control Allocation Framework with SVD-based Protection for a Tilt-rotor VTOL Transition Air Vehicle*
4. *Modeling and Incremental Nonlinear Dynamic Inversion Control for a Highly Redundant Flight System*
5. *Control Allocation Framework for a Tilt-rotor Vertical Take-off and Landing Transition Aircraft*